Multi-Scale LSTM Model with Attention mechanism for short, medium and long term energy forecasting

Shaurya Bhatt
230511815
MSc. Big Data Science(FT)

Abstract*— Energy prediction plays a vital role in the optimization of energy usage in residential buildings, cost reduction, and creating a sustainable environment. Precise predictions ensure smart energy management with the benefits of both economics and environment. This paper employs a Multi-Scale Long Short-Term Memory (LSTM) model to forecast daily, weekly and monthly energy consumption using the Smart Meters London dataset. Owing to their phenomenal property to capture long term dependencies, as opposed to traditional Recurrent Neural Networks (RNNs), LSTMs perform better for energy forecasting. In this paper, we will delve into different LSTM architectures to determine the most effective model for energy forecasting in residential buildings.

Keywords—LSTM, Gated Recurrent Unit, Energy consumption, time-series forecasting

Remove 3rd reference

I. Introduction

Empirical models have demonstrated their ability to effectively forecast energy usage in buildings, but physical models have shown a lack of success in this regard [1]. Subsequently, methods such as Artificial Neural Networks and Support Vector Regression were utilized due to their capability to eliminate structural complexity. A predictive model was developed, taking into account elements such as the types of construction materials and the thicknesses of walls [2]. In Hong Kong, the decision tree model and neural network are the dominant methods for estimating electricity energy usage [3]. Furthermore, feed forward neural networks (FFNNs) demonstrated exceptional accuracy in predicting the daily energy usage for buildings [4]. Recurrent Neural Networks (RNNs) are a specific kind of Artificial Neural Network that incorporate cycles, allowing the network to use earlier time step activations as inputs, which in turn affect predictions at the current time step. The activations are retained in the internal states of the network, enabling RNNs to maintain long-term dependence. RNNs can utilize a dynamic contextual window over the history of the input sequences through this approach. [5]. Traditional recurrent networks are unable to effectively learn when there are time gaps of more than 5 – 10 discrete time steps between relevant inputs and objectives. This is because of the disappearing or exploding gradient problem that occurs during backpropagation through time (BPTT). The problem arises from the phenomenon where an input, as it traverses the network's recurrent connections, has a significant impact on the hidden layer and the network's output, which can either increase exponentially or diminish. To address this constraint, a distinct Recurrent Neural Network (RNN) structure was devised, known as Long Short-Term Memory (LSTM), which has proven to be effective in generating and translating sequences [6,7]. An LSTM layer is comprised of a collection of interconnected blocks, referred to as memory blocks. Each layer consists of several memory cells that are connected in a recurrent manner. Additionally, there are three multiplicative units known as the input, output, and forget gates. These gates perform continuous analogues of write, read, and reset operations for the cells. ... The network can only communicate with the cells through the gates [8]. Therefore, the LSTM architecture effectively solves the issue of the vanishing gradient problem. LSTM models are particularly valuable for tasks that involve sequential data, such as natural language processing and time series prediction. This is because they have the capability to learn long-term dependencies by selectively forgetting or updating information while preserving a cell state. Studies have demonstrated that LSTM models outperform conventional recurrent networks in the field of energy forecasting. Various Long Short-Term Memory (LSTM) models have been suggested as enhancements for energy forecasting, such as the integration of attention mechanisms into LSTM architectures. Similar to their achievements in context-aware applications, these algorithms effectively adjust the relative importance of different temporal stages in the input sequence [9]. Attention-based long short-term memory models (LSTMs) have the ability to effectively capture intricate relationships by giving importance to relevant environmental factors such as weather patterns and historical consumption data. Furthermore, scientific studies indicate that environmental factors, such as temperature, can significantly impact the precision of the proposed model [10,11]. The study aimed to forecast energy usage for two educational buildings by utilizing external characteristics, such as environmental variables, which yielded precise outcomes [12]. Consequently, we have utilized an attention mechanism in addition to incorporating external variables such as weather data, holiday data, and ACORN data to construct a comprehensive forecast model.

The primary purpose of dropout regularization strategies for LSTM in the time domain is to prevent overfitting and improve the overall performance of representatives from the entire class [13,14]. The utilization of these techniques is crucial in energy prediction for the model to maintain its efficacy in the ever-changing environmental and operational conditions. By simplifying the complexity of the model and controlling the Long Short-Term Memory (LSTM) units, we

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

have secured the success of promoting highly reliable energy management techniques in various forecasting time periods.

The research introduces a Multi-Scale Long Short-Term Memory model with attention mechanism for energy forecasting. The model is designed to anticipate energy consumption at several time horizons, specifically daily, weekly, and monthly. The objective is to capture dependencies that occur across short, medium, and long periods of time. The correctness of the model is determined by calculating performance metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). Additionally, various LSTM architectures are tested to select the most optimal one.

Literature Review

Recent research have shown that the implementation of machine learning algorithms has resulted in improved short-term energy consumption predictions by utility providers. Various research publications have utilized methods such as Multilayer Perceptron (MLP), Long Short-Term Memory Support Vector Regression, and gradient boosting approaches like XGBoost, CatBoost, and LightGBM for testing purposes. These methods have proven particularly effective, as they utilized a deliberately designed feature derived from environmental data and past usage trends in the development of each individual one. This enables the generation of more precise short-term hourly projections [15].

A specific study examined the functioning of LSTM models and shown their ability to effectively utilize several intricate aspects in energy data [16]. The primary research verified the significance of LSTMs in learning long-term dependencies. Furthermore, it established that LSTMs are particularly effective in addressing weather-sensitive situations, namely in accurately anticipating long-term weather patterns.

Furthermore, the other study investigated the utilization of machine learning to enhance energy efficiency in houses by manipulating sensors. The energy consumption of households, where sensors were placed, was predicted using regression, decision trees, and random forests. This approach effectively tackled the issues related to cost and data fluctuations [17]. Experienced performers often rely on these widely used approaches for natural learning to evaluate the accuracy of techniques such as LSTM and energy forecasting assignments in the field of energy. Furthermore, data mining has been employed to forecast the monthly energy usage in a building, an overlooked facet of the data's dynamics. This study acknowledges the significance of Convolutional Neural Networks and Feedforward Neural Networks as valuable tools in the field of energy prediction, particularly in the context of deep learning. Building upon this understanding, the investigation also delves into the utilization of LSTM and other sophisticated models.

Furthermore, Deep Neural Networks have been utilized for predicting energy usage over both short and long periods of time. By integrating Recurrent Networks, LSTMs, and Generative Adversarial Networks, it is now possible to generate future patterns of simulated energy use up to two years in advance. This represents a significant advancement in long-term forecast accuracy, as demonstrated by [19]. The available literature overwhelmingly supports the use of LSTM networks for energy prediction.

Comparison between our algorithm and currently available deep learning models

Ref.	Model	Epochs	Hidden neurons	Batch size	Dropout	Value
[23]	LSTM	150	20	-	No	-
[24]	LSTM	150	100	125	No	-
[25]	LSTM	50	30	50	No	-
[26]	LSTM	50	20	1	Yes	0.1
[27]	CNN-LSTM	-	100	-	No	-
[28]	DE-LSTM	3000	50	500	No	-
[29]	LSTM	-	40	-	Yes	0.5

II. METHODOLOGY AND IMPLEMENTATION

The London Smart Meters Dataset offers a comprehensive insight into the energy usage patterns of 5,567 homes in London. The data was included in the Low Carbon London Project of UK Power Networks, and it was collected from November 2011 to February 2014. The dataset has the intention of offering a complete understanding of the usage of energy to back up energy efficiency initiatives to make the smart grid technologies. The readings come from the households are aggregated as daily readings and each house is recognized by means of a unique ID (LCLiD) .The readings are aggregated to daily readings for each house, making the data more manageable. The target variable(energy_mean) is in kilowatt-hours, which is the amount of electricity consumed daily. The dataset comes in the form of a table, with the following columns: Household ID, day,energy_mean, energy_min and other aggregations of the daily energy consumption data.

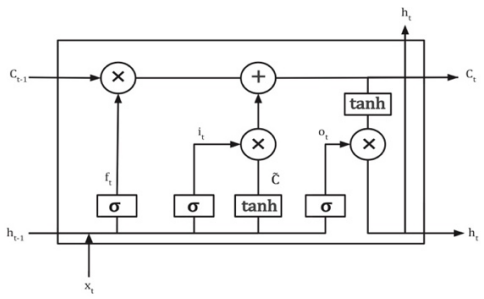


Fig. 1 Structure of Long Short-term Memory (LSTM).

Moreover, the original data will be integrated with external data such as weather data, bank holidays data and household data to build a holistic model. The aim of the study is to build an LSTM model that can capture temporal dependencies in the energy usage data enabling for more accurate predictions.. The model performance is assessed using Mean Absolute Error (MAE) and Root Mean Squared Error(RMSE).

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (2)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (3)$$

$$\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (4)$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \tilde{C}_t \quad (5)$$

$$h_t = o_t \otimes \tanh(C_t) \quad (6)$$

The number of trainable weights in the network are directly proportional to the network's complexity. The connections between the input layer, hidden layer, and output layer, as well as the internal connections of LSTMs, illustrate their functionality. The Number of Trainable Weights (NTW) for a neural network with n inputs, m outputs, and p LSTM cells in the hidden layer can be computed as follows:

$$NTW = 4np + 4pp + 4p + mp + m.$$

This aim of the project was the development of a reliable forecast model that could accurately predict residential energy use using cutting edge machine learning techniques. The objective was to accurately predict energy consumption at multiple temporal scales ranging from daily, weekly, to monthly, by using various inputs such as weather data, data collected from residential houses, and past energy usage trends. The method was structured to capture short-term, medium-term and long-term energy consumption trends. The initial steps were data preprocessing and gathering. The dataset included 111 CSV files containing the unique household ID column, day column and aggregates of the daily energy consumption data like energy_mean,energy_median, etc. The target column was energy_mean. To ensure the dataset's usability, data preprocessing was essential. The first step was to clean and filter the data, which included handling missing values, date string conversions, and filtering the records to include only the information from January 1, 2012, onwards. The files without relevant data columns, such as the 'day' column, were dropped to maintain consistency in the dataset. Feature engineering played an important role in improving the predictive power of the model. To capture temporal patterns in energy consumption, new features such as rolling means over monthly and weekly timeframes were created. These attributes were important in developing the different time scales of energy-use. Moreover, the categorical variables,including holiday and weather data have been transformed into numerical format through one-hot encoding. This was necessary to enable the machine learning model to process categorical data efficiently. Principal Component Analysis (PCA) was employed to reduce the high dimensionality of the dataset and was used to reduce the number of features while preserving 95% of its variance. This led to simplifying the model with minimal loss of information. The PCA explained variance plot shows that there are approximately 120 components that are adequate to capture most of the variance, making the model more manageable.

After dimensionality reduction was made, feature selection was done using Lasso regression. Lasso regression was selected due to it's ability to penalize less important features, thereby selecting only the most relevant predictors. This step resulted in a refined set of features which were most predictive, including various temporal, weather-related, and household features.
The chosen features were used to build a Multi-Scale Long Short Term Memory model with attention mechanisms. The model was developed to predict energy consumption during different timeframes. The input sequences were processed through different LSTM branches, corresponding to daily, weekly, and monthly scales which were used in the model. The attention mechanism was added to the model in order to allow it to prioritise the most important parts of the sequence data, which in turn would improve prediction accuracy. The model was prepared by using the Stochastic Gradient Descent (SGD) optimizer, and the parameters were fine-tuned in such a way that it would balance the learning rate and momentum.

III. IMPLEMENTATION

The language used for coding was Python which used various libraries like pandas, scikit-learn and numpy were used for data processing as well as feature engineering and model development. The data preprocessing pipeline was executed using Pandas. The tasks involved in doing so were filtering of records, reshaping of data and aggregation of energy consumption metrics by day. The model development was employed by using tensorflow. Moreover, the visualisation was done using matplotlib. The merge() function was used to merge external datasets like weather data and bank holidays while ensuring temporal alignment using resampling.
The primary tool used for feature engineering was pandas along with Scikit-learn for tasks such as one-hot encoding. The temporal features were created using the Pandas datetime modules while the lagged features, such as the previous week's total energy, were generated using groupby operations and rolling windows. Multicollinearity between the lagged features was addressed by performing the Variance Inflation Factor (VIF) analysis. The analysis showed high multicollinearity among the lagged features, which led to retain only one lagged feature per time scale. In detail last_3_days_energy_mean, last_3_weeks_energy_mean, and last_3_months_energy_mean were selected to characterize the daily, weekly, and monthly trends, respectively. The VIF values were greatly reduced by this method, enhancing model interpretability and stability. Further feature selection was done using scikit-learn's LassoCV class.

Lasso regression is used to identify and keep the most relevant features by penalising the coefficients of less important factors. A selected features bar plot and the Lasso coefficient path were visualized via Matplotlib which helped in the understanding of regularisation on feature selection..

Dimensionality reduction was accomplished by Principal Component Analysis (PCA) using the PCA class of Scikit-learn. This allowed the model to hold 95% of the variance, allowing reduction in the number of features for less computational complexity. The LSTM model was employed using TensorFlow's Keras API. The data sequences were built with the help of Python functions which was implemented to organize, the scaled input features into 30-day windows, capturing the temporal dependencies within the data. The LSTM architecture included various LSTM layers, along with each one of them corresponding to different forecasting horizons (daily, weekly, and monthly). Attention units were included to boost the model's ability to capture dependencies, which was essential in accurately predicting energy consumption across different time scales. Training of the model was performed using SGD optimizer which was configured with momentum and Nesterov's acceleration to improve convergence. The model was trained on the prepared data with the addition of early stopping to avoid overfitting. Model evaluation encompassed the forecasting of energy consumption across different time scales with the results inverse-transformed for better interpretability. The performance metric used for each forecasting horizon was RMSE, and the training process was visualised using Matplotlib.

Feature Selection and Analysis Methodology

Feature engineering played a crucial role in the project as it affected the predictive power of the model greatly. Hence, various techniques were employed to identify the most influential features.
The initial dataset included a large number of features such as temporal data, weather data and household information. Moreover the categorial features were one hot encoded, expanding the feature space. For example, the 'summary' column in the weather data had unique entries, like "Partly cloudy throughout the day", each such entry was then turned into a binary column 'summary_Partly cloudy throughout the day', with value either 0 or 1, enabling model training. After an exhaustive feature analysis, the final set of features was created by removing about 80 columns. The final feature set has 34 attributes.

The feature analysis included the following:

1. Correlation Analysis

The correlation matrix is a visual representation of the correlation coefficients between a dataset's attributes.
An analysis was performed to check the multicollinearity between features so that among the highly correlated features just one can be included in the feature set. The heat map in Figure-1 provides insight about how the features are linked to itself and other features. The diagonal line represents a particular feature's degree of self-correlation with itself.

Figure-1

The off diagonal values in the matrix represent correlations between different features. The colour scale on the right shows the magnitude and direction of the relationship values. Dark Red stands for a strong positive relationship whereas dark blue demonstrates a negative relationship. The light colours represent weak or non-existent correlation, around 0, i.e no linear correlation.

The main findings were as follows:
1. Heavy Correlation Groupings: From the heatmap, it's evident that Acorn_ACORN-L and Acorn_ACORN are highly correlated features, meaning these features might serve the same purpose and hence only one should be in the final feature set.
The file_block features like File_Block_76, file_block_72, etc., are also in clusters of high correlation. This may be due to the fact that they share some common attributes because of the way they were grouped or treated.

2. Possibility of Repetition:
The strong associations among a number of Acorn and file_block features might indicate the existence of redundancy which can cause overfitting. Taking in account all these similar features would cause the model to be overloaded.

3. Isolated Features:
Is_holiday, temperature, and apparentTemperature emerge as features that are less assorted and less interpolated in the last position, relative to the others. This suggests that they have low multicollinearity and should be added to the final feature set.
4. Multicollinearity:
Multicollinearity happens when two or more features are highly correlated, meaning one can be linearly predicted by the other. This prevents the feature from having any independent effect. Acorns and file_blocks that are highly-correlated can be dropped due to redundancy and only one can be present in the final set.

2.PCA

PCA was utilized in this research to decrease the dataset's dimensionality while preserving the most essential information. The Cumulative Explained volatility (CEV) plot illustrates that the initial principal components effectively capture the majority of the volatility in the data, making them well-suited for constructing a smaller feature

space. The x-axis of the plot reflects the number of primary components, while the y-axis represents the cumulative explained variance. The cumulative explained variance indicates the proportion of the overall data variation that is held by the selected components.

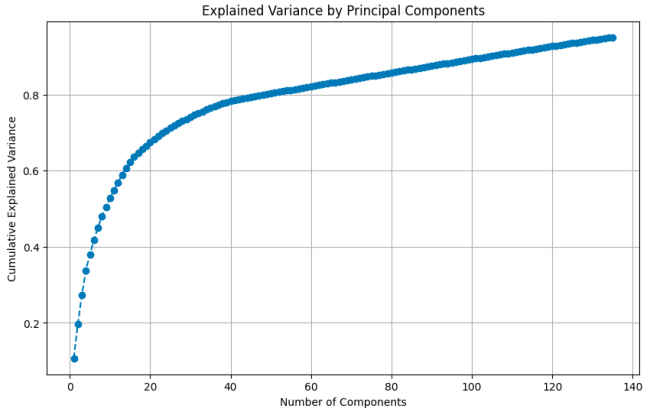


Figure-2

At start, the curve shows a steep increase, suggesting that the initial components contain the majority of the information. As additional components are included, the curve reaches a plateau, indicating that each subsequent component has a diminishing impact on the overall explained variance. Usually, a set of 20-30 components is enough to capture 80-85% of the entire variance. This strikes a reasonable balance between keeping the model simple and retaining a significant amount of volatility. Choosing the optimal number of components requires balancing simplicity with the retention of the most significant variance. For example, selecting 20 components can capture around 80% of the variation, making the model efficient and interpretable. By decreasing the quantity of characteristics to the primary components, the model becomes more efficient and comprehensible, while yet preserving the necessary variation required for precise forecasts.

3. Lasso Regression

Lasso regression, the regularization procedure, which is by default feature selection method that eliminates the irrelevant features by shrinkage, it makes the coefficients zero and thus selects a subset of features. Lasso regression has identified 54 features including weather and holiday features. On the other hand, only those house-related attributes, like Acorn_ACORN-A or stdorToU_ToU, are chosen due to multicollinearity. Both weather conditions and the households' attributes have been considered and have resulted in the final list of numerical and categorical features. The features selected were 'pressure', 'apparentTemperatureMax' , 'month', 'windSpeed', 'last_3_days_energy_mean', 'day_of_week', 'last_3_months_energy_mean', 'last_3_weeks_energy_mean'.

IV. Testing/Evaluation

The dataset is divided into an 80:20 ratio, meaning that 20% of the whole data was used for testing while the remaining data was used for training the model. The model performance was evaluated using multiple metrics and visualisations. Principal Component Analysis (PCA) was tested through the explained variance plot, which revealed that only 120 components were necessary to represent 95% of the variance in the dataset indicating that it efficiently simplified model with minimal loss of information.

The VIF analysis among the selected features was carried out. Only the VIF values for last_3_weeks_energy_mean and last_3_months_energy_mean had little higher values but were considered as they are crucial features for training.

The Lasso regression feature selection process was confirmed by identifying energy consumption's most significant predictors. The selected features served as the basis for the deeper plane of the model and as a result, they made sure that only the most appropriate data were used in the correct model.

The primary metric for the evaluation of the model was RMSE. The RMSE for daily predictions was 0.0911 , for weekly predictions was 0.0848 and for monthly predictions was 0.0842. These RMSE scores indicate a well-functioning model in terms of energy consumption forecasting at different timescales. The performance was assessed through visual comparisons by comparing the actual time series data with the predicted one. The plots showed that the model closely matched the trends although some discrepancies were observed, particularly during rapid change in energy usage, for example the winter months.

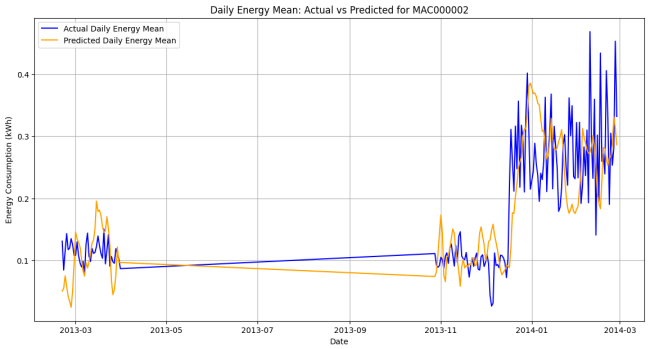
$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|\left(y_{predicted} - y_{actual}\right)\right|$$

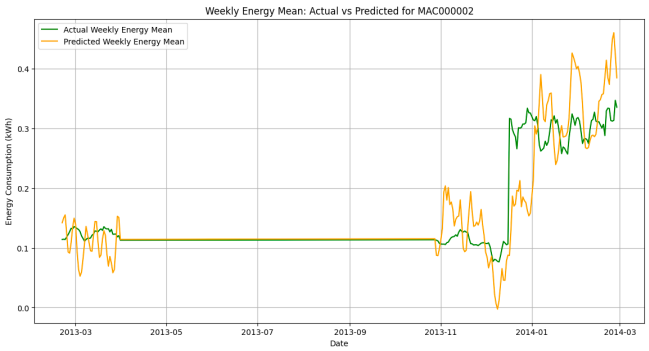
$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_{actual} - y_{predicted}\right)^2}$$

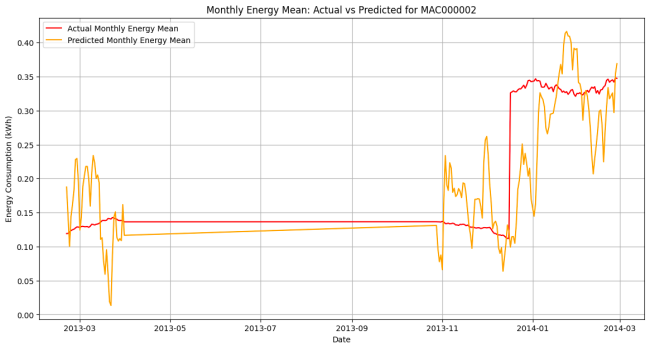
V. Results

The results revealed that the multi-scale LSTM model with attention mechanism outside the household is successfully able to forecast household electricity consumption. The model's ability to integrate temporal, weather, and household data as well as to process those data, enabled the model to capture complex patterns in energy usage. The low RMSE values found in daily, weekly, and monthly predictions were a confirmation of the model's accuracy and robustness. The actual energy use versus the predicted energy use obtained from the images further validated the model's performance. For example, the line diagrams that present the average of daily, weekly, and monthly consumptions of energy of certain households indicated that the model, most of the time, was able to replicate the true variations in the energy consumption data, thus being accurate also in cases of significant fluctuations in energy usage. The results also underscored the importance of feature selection and dimensionality reduction with respect

to building an efficient model. By focusing only on the relevant data inputs, the model was able to achieve good accuracy without overfitting ensuring generalisation.







To conclude, the model was able to develop a robust system for energy forecasting of households in different time stamps. The proposal of applying advanced machine learning techniques such as LSTM with attention mechanisms, together with proper data preprocessing and feature engineering made the project a highly accurate and reliable predictive model. Additional improvements could include fine-tuning the model's hyperparameters and searching for extra variables that might help in increasing its predictive capacity.

VI. Discussion

The multi-scale LSTM model, equipped with attention mechanisms, exhibited a robust capacity to forecast household energy consumption across several temporal intervals, including daily, weekly, and monthly. The technique successfully caught both immediate variations and enduring patterns in energy consumption, which is crucial for applications like predicting demand and managing energy.An important advantage of this strategy was the meticulous feature engineering, which involved creating lagged features that captured past consumption patterns. By specifically examining significant temporal patterns, such as the most recent 3 days, 3 weeks, and 3 months, the model may utilize previous actions to generate precise forecasts for the future. PCA was essential in reducing the dimensionality of the dataset, allowing the model to focus on the most informative characteristics while avoiding noise or redundant data.The utilization of Lasso regression and RandomForestRegressor in the process of feature selection enhanced the model by ensuring the inclusion of just the most pertinent features. This not only strengthened the model's capacity to make accurate predictions, but also improved its interpretability by lowering the presence of multicollinearity and focused on the factors that had the greatest influence on energy usage.The attention mechanism in the LSTM architecture proved highly efficient in recognizing and highlighting the most crucial components of the input sequences. By allowing the model to concentrate on the most pertinent temporal connections, it becomes crucial for achieving precise predictions over various time intervals. The utilization of the SGD optimizer with momentum and Nesterov acceleration facilitated the model's rapid convergence, while early halting mitigated overfitting, yielding a model that exhibits strong generalization to unfamiliar data.Nevertheless, the model displayed certain constraints. The performance, albeit robust, exhibited certain inconsistencies, especially during moments of fast fluctuations in energy use. This suggests that the model might be improved by refining it further, potentially by using more sophisticated structures or by integrating more external elements that may impact energy consumption, such as economic data or social behavior patterns.

VII. Conclusion

The project focused to build a predictive model for household energy consumption through the use of the progressive machine learning techniques, a multi-scale long short-term memory networks incorporated with attention mechanisms, feature engineering, and dimension reduction. The model efficiently combined the data sources, which referred to the weather, households, and temporal features, to predict energy consumption at different time scales. The employment of PCA and Lasso regression for feature selection contributed to the model's enhanced ability. Nevertheless, the model had limitations while dealing with frequent shifts indicating that the model could not capture all the complexities in the data. Furthermore, the model could have been more productive if it had included additional features like socioeconomic and real-time data to show the complexity of energy use patterns.
Even though the project reached its goals more work is needed to improve the model's adaptability for real world applications. Overall, the project depicts a substantial step forward in the field of energy consumption prediction.

VIII. FUTURE WORK

Despite its success in the model, there are possibilities for future exploration and development. One area of study might be improving the model's ability to deal with quick changes in energy consumption. This can be solved by trying architectures like Transformer models that are better than traditional LSTMs. Another scope is adding additional data like economic indicators. Moreover, ensemble learning methods could also be applied. For example, the fusion of the LSTM model with other machine learning models like Gradient Boosting Machines or the unconventional statistical models as a hybrid system so that each method's abilities can be maximized. By working on these areas, future research could improve the usage and quality of the model, hence making it a more efficient tool for energy prediction and regulation.

IX. REFERENCES

1. Ahmad, A.S., Hassan, M.Y., Abdullah, M.P., Rahman, H.A., Hussin, F., Abdullah, H., Saidur, R., 2014. A review on applications of ANN and SVM for building electrical energy consumption forecasting. *Renewable and Sustainable Energy Reviews*, 33, pp.102–109.

2. Tso, G.K.F. and Yau, K.K.W., 2007. Predicting electricity energy consumption: A comparison of regression analysis, decision tree and neural networks. *Energy*, 32(9), pp.1761–1768.

3. Kong, W., Dong, Z.Y., Jia, Y., Hill, D.J., Xu, Y. and Zhang, Y., 2019. Short-Term Residential Load Forecasting Based on LSTM Recurrent Neural Network. *IEEE Transactions on Smart Grid*, 10(1), pp.841–851.

4. Qing, X. and Niu, Y., 2018. Hourly day-ahead solar irradiance prediction using weather forecasts by LSTM. *Energy*, 148, pp.461–468.

5. Sak, H., Senior, A. and Beaufays, F., 2014. Long Short-Term Memory Recurrent Neural Network Architectures for Large Scale Acoustic Modeling. *arXiv preprint arXiv:1402.1128*.

6. Graves, A., Mohamed, A.R. and Hinton, G., 2005. Framewise Phoneme Classification with Bidirectional LSTM and Other Neural Network Architectures. *Neural Networks*, 18(5-6), pp.602–610.

7. Hochreiter, S. and Schmidhuber, J., 1997. Long Short-Term Memory. *Neural Computation*, 9(8), pp.1735–1780.

8. Bahdanau, D., Cho, K. and Bengio, Y., 2015. Neural machine translation by jointly learning to align and translate. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

9. Hong, T., Pinson, P., Fan, S. and Zareipour, H., 2020. Probabilistic load forecasting: A tutorial review. *International Journal of Forecasting*, 36(3), pp.938–965.

10. Wang, Z., Wang, Y., Zeng, R., Srinivasan, R.S. and Ahrentzen, S., 2018. Random Forest based hourly building energy prediction. *Energy and Buildings*, 171, pp.11–25.

11. Kong, W., Dong, Z.Y., Jia, Y., Hill, D.J., Xu, Y. and Zhang, Y., 2019. Short-Term Residential Load Forecasting Based on LSTM Recurrent Neural Network. *IEEE Transactions on Smart Grid*, 10(1), pp.841–851.

12. Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I. and Salakhutdinov, R., 2014. Dropout: A simple way to prevent neural networks from overfitting. *The Journal of Machine Learning Research*, 15(1), pp.1929–1958.

13. Runge, J. and Zmeureanu, R., 2021. A Review of Deep Learning Techniques for Forecasting Energy Use in Buildings. *Energies*, 14(10), p.2883.

14. Wong, A., Lai, Y., Jow, T., Wang, C., Liu, C. and Cheng, M., 2022. Machine Learning Models Application in Daily Forecasting of Hourly Electricity Usage. In *2022 IEEE International Systems Conference (SysCon)*, Montreal, QC, Canada, pp.1–5. doi: 10.1109/SysCon53536.2022.9773835.

15. Thakur, A., Shukla, K.A., Choudhary, A. and Atrey, J., 2023. Predictive Analysis of Energy Consumption and Electricity Demand Using Machine Learning Techniques. In *2023 International Conference on Smart Systems for Applications in Electrical Sciences (ICSSES)*, Tumakuru, India, pp.1–6. doi: 10.1109/ICSSES58299.2023.10200636.

16. Edwards, R.E., New, J. and Parker, L.E., 2012. Predicting future hourly residential electrical consumption: A machine learning case study. *Energy and Buildings*, 49, pp.591–603.

17. Akter, R., Lee, J.-M. and Kim, D.-S., 2021. Analysis and Prediction of Hourly Energy Consumption Based on Long Short-Term Memory Neural Network. In *2021 International Conference on Information Networking (ICOIN)*, Jeju Island, Korea (South), pp.732–734. doi: 10.1109/ICOIN50884.2021.9333968.